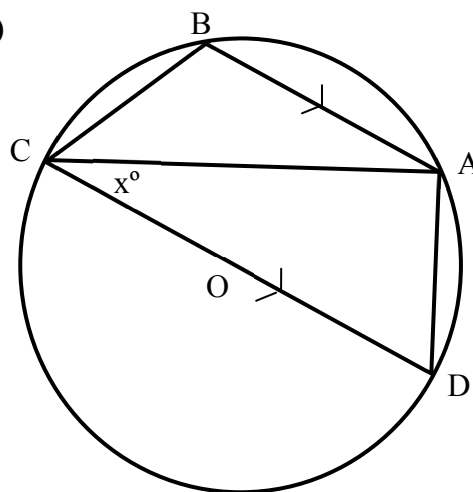


QUESTION 1. (Start on a new sheet of paper)**MARKS**

- a) If $y = x \tan^{-1} x$, find $\frac{dy}{dx}$. 2
- b) If $f(x) = \sin^{-1}(1 - 2x)$, show that $f'(x) = \frac{-1}{\sqrt{x - x^2}}$. 3
- c) $P(x)$ is an odd polynomial of degree 3. It has $(x + 4)$ as a factor and, when it is divided by $(x - 3)$, the remainder is 21. Find $P(x)$. 3
- d) By making the substitution $u = x - 2$, evaluate $\int_4^5 \frac{x(x-4)}{(x-2)} dx$ 4

QUESTION 2. (Start on a new sheet of paper)

- a) The points A, B, C and D lie on the circumference of a circle centred at O . CD is a diameter of the circle and AB is parallel to CD . $\angle ACD = x^\circ$. Find an expression for $\angle ACB$ in terms of x . 3

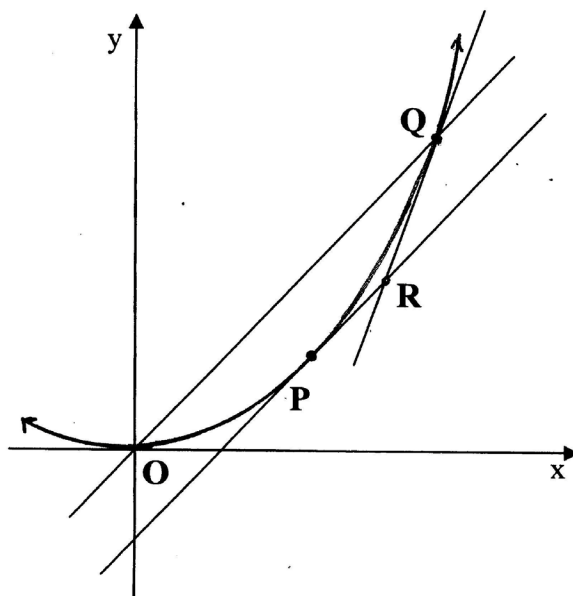


- b) Use the method of mathematical induction to show that the expression $9^n - 8n - 1$ is divisible by 64 for all integers $n \geq 2$. 5
- c) i) Given that ${}^nC_r = \frac{n!}{r!(n-r)!}$, show that $\frac{r \times {}^nC_r}{{}^nC_{r-1}} = n - r + 1$ 1
- ii) Hence show that : 3
- $$\frac{{}^nC_1}{{}^nC_0} + \frac{2 \times {}^nC_2}{{}^nC_1} + \frac{3 \times {}^nC_3}{{}^nC_2} + \dots + \frac{n \times {}^nC_n}{{}^nC_{n-1}} = \frac{n}{2}(n+1).$$

QUESTION 3. (Start on a new sheet of paper)**MARKS**

a) Evaluate the definite integral $\int_0^{\frac{1}{\sqrt{2}}} \frac{x}{\sqrt{1-x^4}} dx$ by using the substitution $u = x^2$. **4**

- b) The point $P(2ap, ap^2)$ is on the parabola $x^2 = 4ay$ and a straight line OQ is drawn through the vertex parallel to the tangent at P . This line meets the parabola again at Q and the tangent to the parabola at Q meets the tangent at P in R , as shown in the diagram.



You are given that the tangent at P has equation $y = px - ap^2$.

- i) Write down the equation of the line OQ . **1**
- ii) Find the coordinates of Q in terms of a and p . **2**
- iii) Show that the equation of the tangent at Q is $y = 2px - 4ap^2$. **1**
- iv) Find the coordinates of R . **2**
- v) Show that, as P varies on the parabola, R moves on another parabola whose equation is $x^2 = \frac{9}{2}ay$. **2**

QUESTION 4. (Start on a new sheet of paper)**MARKS**

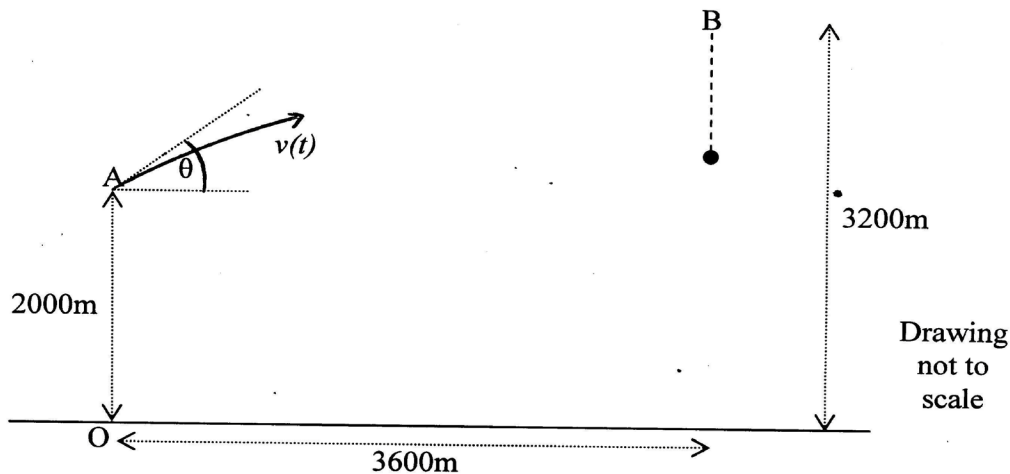
- a) Consider the function $f(x) = \frac{e^x}{(1 + e^x)}$.
- i) Find $f'(x)$ and deduce that $f(x)$ is increasing for all x . 2
 - ii) State the range of $f(x)$. 1
 - iii) Find the inverse function $f^{-1}(x)$ 2
 - iv) Draw $y = f(x)$ and $y = f^{-1}(x)$ on the same diagram. 2
- b) A particle moves in a straight line on the x axis. At time t its velocity is v and its acceleration is a .
- i) If $a = 4x - 4$ and initially $x = 6$ and $v^2 = 64$, show that $v^2 = 4x^2 - 8x - 32$. 2
 - ii) Use this expression for v^2 to find the possible values of x . 1
 - iii) Describe the motion of the particle if $v = -8$ initially. 2

QUESTION 5. (Start on a new sheet of paper)

- a) A particle moves in a straight line with displacement in centimetres from the point $x = 0$ at time t seconds given by $x = \sin 3t + 2 \cos 3t$ for $t \geq 0$.
- i) Express x in the form $R \sin(3t + \alpha)$ where $R > 0$ and $0 \leq \alpha \leq \frac{\pi}{2}$. 2
 - ii) Show that the motion is simple harmonic. 1
 - iii) Write down the period of motion. 1
 - iv) Find at what time, to the nearest tenth of a second, the particle first reaches $x = 1$. 2
- b) Fred deposited \$20,000 at the beginning of January into an account which paid interest at the rate of 0.5% per month compounded monthly. He withdrew \$50 each month from the account each month, immediately after the interest was paid.
- i) How much money was in the account immediately after the first withdrawal? 1
 - ii) Show that, after making the n^{th} withdrawal, his account balance is given by the expression $\$(10,000 \times 1.005^n + 10,000)$ 3
 - iii) Find the number of months it will take for his account balance to be \$50,000 2

QUESTION 6. (Start on a new sheet of paper)**MARKS**

- a) Find the term independent of x in the expansion of $\left(\frac{2x^2}{3} - \frac{3}{2x}\right)^9$. **3**
- b) An aeroplane, A , flying at a height of 2000m observes a stationary blimp, B , at a height of 3200m drop an object. As the object is dropped, the plane fires a projectile towards it at a speed of 240m/s and at an angle θ to the horizontal. The horizontal distance between the plane and the blimp is 3600m at the time that the projectile is fired.



The origin of coordinates, O , is taken to be the point on the ground below A .

The particle's coordinates at time t (secs) are given by :

$$x = 240t \cos \theta,$$

$$y = 2000 + 240t \sin \theta - \frac{gt^2}{2}$$

The coordinates of the dropped object at time t are :

$$x = 3600,$$

$$y = 3200 - \frac{gt^2}{2}$$

(You may use $g=10\text{m/s}^2$)

- i) What is the angle θ at which the projectile must be fired to intercept the object, and how long does it take to reach it? **3**
- ii) At what height does the projectile intercept the object? **1**
- c) A man notices two towers, one due North and one in a direction $N\theta E$ (i.e. at an angle θ east of north). The angle of elevation β of both towers is the same but the height of one tower is twice the height of the other. Show that

$$\cos \theta = \frac{5 \cot^2 \beta - \cot^2 \alpha}{4 \cot^2 \beta} \quad \text{5}$$

where α is the angle of elevation of the top of the taller tower from the top of the shorter.

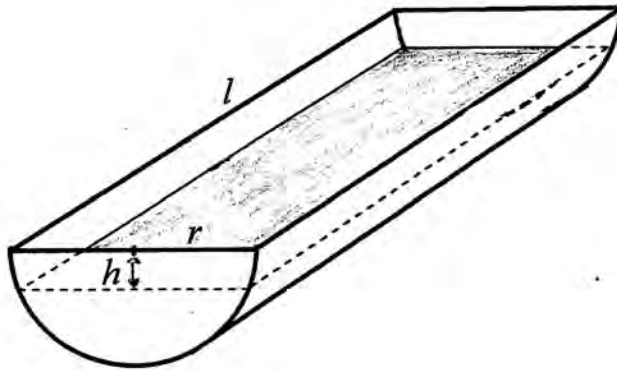
QUESTION 7. (Start on a new sheet of paper)**MARKS**

- a) A group of four contestants for a quiz game are to be selected at random from a class of eight girls and five boys.

i) What is the probability that the team comprises three girls and one boy? **2**

ii) Find the probability that there are more girls chosen than boys. **2**

- b) A water trough takes the shape of a hollow semi-circular prism with length l and radius r . It is placed on horizontal ground and filled with water. The surface of the water is at a distance h below the top of the trough, as shown in the diagram.



- i) Show that the area A of the flat surface area of water is given by

$$A = 2l\sqrt{r^2 - h^2} \quad \mathbf{2}$$

- ii) Show that the volume V of water in the trough is given by

$$V = l \left(r^2 \cos^{-1} \left(\frac{h}{r} \right) - h\sqrt{r^2 - h^2} \right) \quad \mathbf{2}$$

- iii) If the water level is falling, show that $\frac{dV}{dt} = -2l\sqrt{r^2 - h^2} \frac{dh}{dt} = -A \frac{dh}{dt}$. **3**

- iv) On a sunny day, the rate of evaporation at any time (and hence $-\frac{dV}{dt}$) is proportional to A . Show that the water level falls at a constant rate. **1**

END OF THE PAPER

1) a) $y = x \tan^{-1} x$
 $\frac{dy}{dx} = \tan^{-1} x + \frac{x}{1+x^2}$ (Product rule)

b) $f(x) = \sin^{-1}(1-2x)$
 Let $u = 1-2x$, $\frac{du}{dx} = -2$,
 $\therefore f'(x) = \frac{1}{\sqrt{1-(1-2x)^2}} \cdot x - 2$
 $= \frac{-2}{\sqrt{1-(1+4x^2-4x)}} = \frac{-2}{\sqrt{4x-4x^2}} = \frac{-1}{\sqrt{x-x^2}}$

c) $P(x) = Ax(x+4)(x-4)$
 $P(3) = 3A \times 7 \times -1 = 21 \therefore A = 1$
 $\therefore P(x) = -x(x+4)(x-4)$
 $(= 16x - x^3)$

1) Let $I = \int_4^5 \frac{x(x-4)}{(x-2)} dx$
 Let $u = x-2$ $x=5 \Rightarrow u=3$
 $x=4 \Rightarrow u=2$
 $\frac{du}{dx} = 1 \therefore "dx = du"$
 $\therefore I = \int_2^3 \frac{(u+2)(u-2)}{u} du$
 $= \int_2^3 \frac{u^2-4}{u} du$
 $= \int_2^3 u - \frac{4}{u} du$
 $= \left[\frac{u^2}{2} - 4 \ln u \right]_2^3$
 $= \left(\frac{9}{2} - 4 \ln 3 \right) - \left(2 - 4 \ln 2 \right)$
 $= \frac{5}{2} - 4 \ln \left(\frac{3}{2} \right)$

2) a) ABCD is a cyclic quadrilateral
 $\angle CAD = 90^\circ$ (Angle at circumference in semi circle)
 $\angle BAC = x^\circ$ (Alternate angles are equal $AB \parallel CD$)
 $\therefore \angle BAD = x^\circ + 90^\circ$
 $\therefore \angle BCD = 180^\circ - (x^\circ + 90^\circ)$ (Opposite angles in cyclic quad. supp.)
 $= 90^\circ - x^\circ$
 $\therefore \angle BCA = \angle BCD - \angle ACD$
 $= 90^\circ - 2x^\circ$

b) When $n=2$,
 $T_2 = 9^2 - 16 - 1 = 64$
 which is divisible by 64
 So induction starts.
 Assume that
 $9^k - 8k - 1 = 64A$ for $k \geq 2$
 and $A \in \mathbb{Z}$.

Then
 $9^{k+1} - 8(k+1) - 1 = 9 \cdot 9^k - 8k - 9$
 $= 9(9^k - 1) - 8k$
 $= 9(9^k - 8k - 1) + 64k$
 $= 9 \times 64A + 64k$
 $= 64(9A + k)$
 Thus if true for $n=k$, also true for $n=k+1$.
 $\therefore$ By principle of M.I.,
 $9^n - 8n - 1$ is divisible by 64 for $n \geq 2$

c) i) $\frac{{}^nC_r}{{}^nC_{r-1}} = \frac{r!}{(n-r)!r!} \cdot \frac{(n-r)!}{n!} \cdot \frac{(n-r)!}{(n-r)!}$
 $= \frac{r(r-1)!}{r!} \cdot \frac{(n-r)!}{(n-r)!}$
 $= \frac{n-r+1}{1}$

2 c) ii) Using result from 1)
 $n + (n-1) + (n-2) + \dots + 2 + 1$
 This an AP with $a=n, d=-1$
 $\text{Sum} = \frac{n}{2}(2n + (n-1)(-1))$
 $= \frac{n(n+1)}{2}$

3) a) Let $I = \int_0^{\sqrt{2}} \frac{x dx}{\sqrt{1-x^4}}$
 Let $u = x^2$ $x = \sqrt{2} \Rightarrow u = 1$
 $"du = 2x dx"$ $x=0 \Rightarrow u=0$
 $\therefore I = \frac{1}{2} \int_0^1 \frac{du}{\sqrt{1-u^2}}$
 $= \left[\frac{1}{2} \sin^{-1} u \right]_0^1$
 $= \frac{1}{2} \cdot \frac{\pi}{6} = \frac{\pi}{12}$

b) i) $y = px$
 ii) Crosses parabola where
 $x^2 = 4apx$
 $x = 4ap$ ($x \neq 0$)
 $y = 4ap^2$
 Q is $(4ap, 4ap^2)$

iii) Q is point with parameter " $2p$ "
 $\therefore$ Tangent is $y = (2p)x - a(2p)^2$
 $y = 2px - 4ap^2$

iv) Two tangents cross at R
 $px - ap^2 = 2px - 4ap^2$
 $\Rightarrow x = 3ap$ ($p \neq 0$)
 $y = 2ap^2$
 R is $(3ap, 2ap^2)$

v) Eliminate p from $x = 3ap$
 $y = 2ap^2$
 $R = \frac{x}{3a}, y = 2a \left(\frac{x}{3a} \right)^2$
 $x^2 = \frac{9ay}{2}$

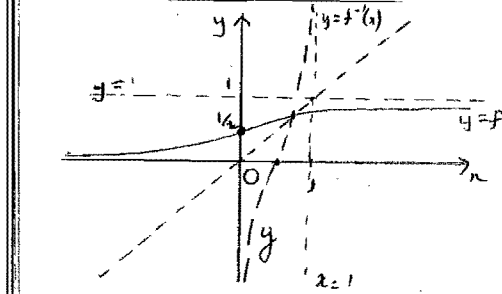
1P. 0.1. 0.4. 0.6. 0.8. 1.0. 1.2. 1.4. 1.6. 1.8. 2.0.

(4) i) $f(x) = 1 - \frac{1}{1+e^x} = 1 - (1+e^x)^{-1}$
 $f'(x) = \frac{e^x}{(1+e^x)^2}$

$e^x > 0$ $(1+e^x)^2 > 0 \therefore f'(x) > 0$
 $\therefore f(x)$ increasing for all x

ii) Range $\{y: 0 < y < 1\}$

iii) Let $y = 1 - \frac{1}{1+e^x}$
 $\frac{1}{1+e^x} = 1-y$
 $1+e^x = \frac{1}{1-y}$
 $e^x = \frac{1}{1-y} - 1 = \frac{y}{1-y}$
 $f'(x) = \ln \left(\frac{y}{1-y} \right)$



b) i) $\frac{d}{dx} \left(\frac{v^2}{2} \right) = 4x - 4$
 $\therefore \frac{v^2}{2} = 2x^2 - 4x + k$
 When $x=6$, $v^2=64$
 $\therefore 32 = 72 - 24 + k \Rightarrow k = -16$
 $\therefore v^2 = 4x^2 - 8x - 32$
 ii) $v^2 = 4(x^2 - 2x - 8)$
 $= 4(x-4)(x+2)$
 $v^2 \geq 0 \therefore -2 \leq x \leq 4$

iii) Particle moving to left from $x=6$
 Stops at $x=4$ and immediately moves off to right, accelerating
 $\therefore$ initial velocity

Equating coefficients:

$$\begin{cases} R \sin \alpha = 2 \\ R \cos \alpha = 1 \end{cases} \quad R = \sqrt{5} \quad \tan \alpha = 2$$

$$\therefore x = \sqrt{5} \sin(3t + \tan^{-1}(2))$$

$$\text{ii) } \ddot{x} = 3\sqrt{5} \cos(3t + \tan^{-1}(2))$$

$$\ddot{x} = -9\sqrt{5} \sin(3t + \tan^{-1}(2))$$

$$\ddot{x} = -9x$$

which is of the form $\ddot{x} = -\omega^2 x$
which defines S.H.M.

$$\text{iii) Period} = \left(\frac{2\pi}{\omega}\right) = \frac{2\pi}{3} \text{ secs.}$$

iv) When $x = 1$,

$$\sin(3t + \tan^{-1}(2)) = \frac{1}{\sqrt{5}}$$

$$3t + \tan^{-1}(2) = n\pi + (-1)^n \sin^{-1}\left(\frac{1}{\sqrt{5}}\right)$$

$n=1$ gives 1st +ve sol:

$$3t = \pi - \sin^{-1}\left(\frac{1}{\sqrt{5}}\right) - \tan^{-1}(2)$$

$$t = \frac{1}{3} \left(\pi - \sin^{-1}\left(\frac{1}{\sqrt{5}}\right) - \tan^{-1}(2) \right)$$

$$= 0.5 \text{ secs (to 10)}$$

b) i) after 1 withdrawal

$$\$ (20000 \times 1.005 - 50) = \$20050$$

ii) Let $P = \$20000$, $D = \$50$, $r = 0.005$

$$\text{After 1 month } P(1+r) - D$$

$$\therefore 2 \dots P(1+r)^2 - D(1+r) - D$$

$$\therefore n \dots P(1+r)^n - D(1+r)^{n-1} - D(1+r)^{n-2} - \dots - D$$

$$= \frac{P(1+r)^n - D((1+r)^n - 1)}{r}$$

$$= 20000(1.005)^n - 10000(1.005)^n + 10000$$

$$= 10000(1.005)^n + 10000$$

$$10000 \times (1.005)^n \geq 40000$$

$$(1.005)^n \geq 4$$

$$n \log(1.005) \geq \log 4$$

$$n \geq \frac{\log 4}{\log 1.005}$$

$$\therefore n = 278 \text{ (months)}$$

$$\text{⑥ a) } r^{\text{th}} \text{ term is } {}^nC_r \left(\frac{2x}{3}\right)^r \left(\frac{-3}{2x}\right)^{9-r}$$

$$= {}^nC_r \left(\frac{2}{3}\right)^r \left(\frac{-3}{2}\right)^{9-r} x^{3r-9}$$

$$\text{Required term has } 3r-9=0 \quad r=3$$

$$\therefore \text{Coefficient is } {}^nC_3 (-1)^4 \left(\frac{3}{2}\right)^3$$

$$= {}^nC_3 \left(\frac{3}{2}\right)^3 = \frac{567}{2} = 283\frac{1}{2}$$

b) i) Intercept when x and y equal simultaneously

$$240t \cos \theta = 3600$$

$$t \cos \theta = 15$$

$$t = 15 \sec \theta$$

$$2000 + 240t \sin \theta - \frac{gt^2}{2} = 3200 - \frac{gt^2}{2}$$

$$240t \sin \theta = 1200$$

$$t \sin \theta = 5$$

$$\text{Solving } \tan \theta = \frac{1}{3}$$

$$\theta = \tan^{-1}\left(\frac{1}{3}\right) = 18^\circ 26'$$

$$\text{At time } t = 15 \sec \theta$$

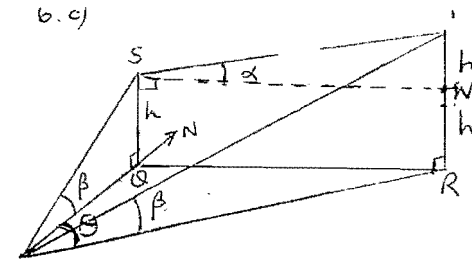
$$= 5\sqrt{10} \text{ sec. } (= 15.81 \text{ s})$$

ii) Putting $t = 5\sqrt{10}$ and $g = 10$ into last equation

$$\text{height} = 3200 - \frac{10 \cdot 250}{2}$$

$$= 1950 \text{ m}$$

b. c)



WPQRST defined in diagram

$\triangle PSQ \parallel \triangle PTR$ (Equiangular)

$$\therefore \frac{PR}{PQ} = \frac{RT}{QS} = 2$$

$$\text{But } PQ = h \cot \beta$$

$$\therefore PR = 2h \cot \beta$$

$$QR = SW = h \cot \alpha$$

Apply cosine rule to $\triangle PQR$

$$\cos \theta = \frac{PQ^2 + PR^2 - QR^2}{2 \cdot PQ \cdot PR}$$

$$= \frac{h^2 \cot^2 \beta + 4h^2 \cot^2 \beta - h^2 \cot^2 \alpha}{2 \cdot h \cot \beta \cdot 2h \cot \beta}$$

$$\cos \theta = \frac{5 \cot^2 \beta - \cot^2 \alpha}{4 \cot^2 \beta}$$

⑦ a) i) Ways of choosing team

$$\text{of } 4 = {}^3C_4 (= 715)$$

$$\text{Ways of choosing 3}$$

$$\text{boys and 1 girl } = {}^8C_3 {}^5C_1$$

$$= 280$$

$$\therefore \text{Probability} = \frac{{}^8C_3 {}^5C_1}{{}^{13}C_4} (= 0.39)$$

ii) Ways of choosing 4 girls

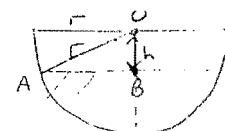
$$= {}^8C_4 = 70$$

$$\therefore \text{Prob all girls } = \frac{{}^8C_4}{{}^{13}C_4}$$

Prob 3 or 4 girls

$$= \frac{{}^8C_4 + {}^8C_3 {}^5C_1}{{}^{13}C_4} (= 0.49)$$

b)



i) Consider cross section of trough.

$$OA = r \therefore AB = \sqrt{r^2 - h^2} \text{ (Pythag)}$$

$$\therefore \text{Surface area} = l \times 2AB = 2l\sqrt{r^2 - h^2}$$

ii) Let $\angle AOB = \theta (= \cos^{-1}(\frac{h}{r}))$

Area of shaded segment is

$$\frac{r^2}{2} (2\theta - \sin 2\theta)$$

$$= \frac{r^2}{2} (\theta - \sin \theta \cos \theta)$$

$$= r^2 \left(\cos^{-1}\left(\frac{h}{r}\right) - \frac{\sqrt{r^2 - h^2}}{r} \cdot \frac{h}{r} \right)$$

$$\therefore V = l \left(r^2 \cos^{-1}\left(\frac{h}{r}\right) - h \sqrt{r^2 - h^2} \right)$$

$$\text{iii) } \frac{dV}{dt} = l \left(\frac{-\frac{h}{r}}{\sqrt{1 - \frac{h^2}{r^2}}} - \sqrt{r^2 - h^2} \right) \frac{dh}{dt}$$

$$+ \frac{h^2}{\sqrt{r^2 - h^2}} \frac{dh}{dt}$$

$$= \left(\frac{-r^2 - (r^2 - h^2) + h^2}{\sqrt{r^2 - h^2}} \right) l \frac{dh}{dt}$$

$$= \frac{-2l(r^2 - h^2)}{\sqrt{r^2 - h^2}} \frac{dh}{dt}$$

$$= -2l \sqrt{r^2 - h^2} \frac{dh}{dt} = -A \frac{dh}{dt}$$

$$\text{iv) } -\frac{dV}{dt} < A \therefore \frac{dV}{dt} = kA$$

where k is constant of proportionality

$$\therefore kA = A \frac{dh}{dt} \Rightarrow \frac{dh}{dt} = k$$

$\therefore h$ increasing at constant rate
(i.e. water falls at constant rate)